MIL-R-0052499D(ME)
31 March 1986
USED-IN-LIEU-OF
MIL-R-52499C(ME)
8 February 1983

MILITARY SPECIFICATION

ROLLERS, TOWED, SMOOTH DRUM, VIBRATORY, AIRBORNE/AIRMOBILE, SINGLE-DRUM

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a diesel-engine-powered, towed-type, smooth-drum, vibratory roller.

1.2 Classification. Rollers shall be one of the following types, as specified (see 6.2):

Type I - 30-inch drum diameter.
Type II - 48-inch drum diameter.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

VV-F-800 - Fuel Oil Diesel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development and Engineering Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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ZZ-B-225 - Belts, V: Industrial (Multiple Drive).

MILITARY

MIL-V-173 - Varnish, Moisture-and-Fungus Resistant, (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-L-2104 - Lubricating Oil, Internal Combustion Engine Tactical Service.
MIL-R-3075 - Rollers, Motorized, Road, Diesel - or Gasoline-Engine-Driven, Packaging of.
MIL-G-3859 - Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-A-8421 - Air Transportability Requirements, General Specification for.
MIL-G-10924 - Grease, Automotive and Artillery.
MIL-G-23827 - Grease, Aircraft and Instrument, Gear and Actuator Screw, Nato Code Number G-354, Metric.
MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-T-52510 - Tractor, Full Tracked, Low-Speed, Diesel Engine-Driven, Air Transportable, Air-Droppable and Helicopter Transportable.

STANDARD

MILITARY

MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down of Military Equipment.
MIL-STD-669 - Loading Environment and Related Requirements for Platforms Rigged Airdrop Materiel.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-814 - Requirements for Tiedown, Suspension and Extraction Provisions on Military Materiel Airdrop.
MIL-STD-838 - Lubrication of Military Equipment.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment, and Facilities.
MIL-STD-1474 - Noise Limits for Army Materiel.
MS35000 - Battery, Storage, Lead-Acid, Waterproof.
MS35489 - Grommets, Synthetic and Silicone Rubber, Hot Oil and Coolant Resistant.
MS35643 - Gasket: Synthetic Rubber, Cap, Fuel Tank.

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MS35644	- Filler Neck: Fuel Tank-Military Vehicles.
MS35645	- Cap, Fuel Tank: Tactical Vehicle, Fording.
MS53063	- Indicator, Air Cleaner, Intake; Restriction Mechanical Type.
MS52131	- Connector, Plug, Electrical Intervehicular Power Cable.

HANDBOOKS

MILITARY

MIL-HDBK-113	- Guide for the Selection of Lubricants, Functional Fluids, Preservatives and Speciality Products for use in Ground Equipment Systems.

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practices for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

B2.1 - Standard for Welding Procedure and Performance Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 Northwest Seventh Street, Miami, FL 33125.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Nongovenment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The smooth drum, vibratory, airmobile, single drum, towed roller (hereinafter called "roller") shall consist of a vibrating mechanism contained within a drum, a frame, a front drawbar with towing hitch, a rear clevis, replaceable drum cleaners, diesel engine, vibrating mechanism and remote control.

3.1.1 Performance. The roller shall withstand rough operation encountered when towed over loose fill dirt, excavation areas, and rock fill areas. The roller shall be capable of performing at its rated capability when towed by crawler tractors, rubber-tired tractors or trucks (as required). The roller shall be capable of performing as specified herein after being transported by truck, trailer, rail, ocean-going carriers, or as an internal or slung load from helicopter organic to airmobile military operations.

3.1.2 Dimensions and weight. The type I and type II rollers shall conform to the requirements listed in table I.

TABLE I. Dimensions, weight and dynamic force.

Requirements	Type I	Type II
Overall height, maximum	72 in.	72 in.
Overall width, maximum	68 in.	88 in.
Drum diameter	28 to 32 in.	48 to 52 in.
Drum width	54 to 60 in.	58 to 66 in.
Drum shell and head thickness, minimum.	5/8 in.	3/4 in.
Empty weight, maximum	5000 lb.	10,000 lb.
Centrifugal force, minimum	10,000 lb.	18,000 lb.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of rollers from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish rollers conforming to this specification.

3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials and special treatments utilized shall provide

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protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the item may be exposed.

3.3.1 Dissimilar metals. Dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact without suitable protection in order to prevent or mimimize galvanic corrosion.

3.3.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and subcomponents, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the rollers may be newly fabricated from recovered materials to the maximum extent practicable, provided the rollers produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the rollers.

3.4 Safety. All rotating or reciprocating parts, and all parts subject to high operational temperature that are of such nature or are so located as to be, or become, a hazard to operating or maintenance personnel shall be guarded or insulated to the extent necessary to eliminate the hazard. All platforms, steps, and ladder rungs shall have antiskid surfaces. All exhausts shall be directed so as not to endanger personnel or obstruct the view of the operator. Fuel tanks shall be located in a manner which will not allow spills and overflows to run onto the engine, exhaust, or electrical equipment. Electrical equipment shall be insulated, fully enclosed, or guarded. Nonfunctional sharp edges, projecting points and protruding length more than two full threads of fastening devices shall be avoided.

3.5 Human factors engineering. The roller shall conform to accepted human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to, MIL-STD-1472 General Requirements (4), Control/Display Integration (5.1), Visual Displays (5.2), Controls (5.2), Controls (5.4), Labeling (5.5), Anthropometry (5.6), Design for Maintainability (5.9), Design of Equipment for Remote Handling (5.10) and Hazards and Safety (5.13), as applicable.

3.5.1 Noise limits. The noise produced by the roller shall conform with MIL-STD-1474 requirements when tested in accordance with 4.5.2.12. The provisions of MIL-STD-1474, 4.3 and 4.4 shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85 dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474, 4.3 and be readable at the distance indicated on the sign.

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3.6 Maintenance.

3.6.1 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement with minimum removal of other major assemblies and installed attachments. Covers or plates which must be removed for component adjustment or maintenance shall be equipped with quick-disconnect fastenings. Drain outlets shall be accessible. Ease of maintenance provisions shall incorporate features insuring operating clearances for handtools and lubricating equipment, and the accessibility for service operations such as adding oil to the engine crankcase and adding oil to gear cases. Each maintenance, assembly, or disassembly operation required during testing in accordance with 4.3 shall be accomplished by no more than two men (except a hoist and operator may be used as necessary) using special tools furnished with the roller and common tools. After completion of any maintenance operation, the roller shall operate as specified herein.

3.6.2 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Common hardware and fastening devices such as bolts, nuts, screws, and washers shall be held to a minimum number of types and sizes.

3.7 Frame. The frame shall be constructed of structural steel shapes or plates. The frame shall be isolated from the drum and vibrating mechanism. The frame shall incorporate a drawbar and adjustable retractable support leg.

3.7.1 Drawbar. The drawbar shall permit pulling and pushing the roller. The roller shall be capable of being coupled for towing after being stabilized in parking position by the support leg (see 3.7.3). Unless otherwise specified, the drawbar shall be equipped with a lunette ring. The lunette ring shall conform to SAE J847. The drawbar lunette ring mounting shall provide a lunette ring height range from 27 to 33 inches.

3.7.2 Crawler tractor drawbar. When specified (see 6.2) the roller drawbar and lunette ring shall be compatible with a crawler tractor drawbar conforming to SAE J749a, pin size 1.75.

3.7.3 Support leg. An adjustable support leg shall be mounted on the drawbar. The support leg shall be vertically adjustable for not less than 12 inches. The support leg shall either retract or be hinged and fold to provide not less than 10 inches ground clearance. The support leg shall be arranged for positive locking into park and operational (travel) position. The support leg shall incorporate a self-alining pad which will provide a ground bearing load not greater than 20 pounds per square inch when the maximum drawbar weight is imposed. The edges of the pad shall be turned up.

3.8 Drum. The shell and heads of the drum shall be constructed of steel plates. The outer surface of the shell shall be smooth. The joints between shell and heads shall be welded and the heads shall not incorporate provisions for ballasting.

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3.9 Axles and bearings. The drum shall be supported to provide independent rolling capabilities. The vibrating shaft shall extend entirely through the drum. The vibrating shaft may act as a drum axle through the use of a hollow collar connected to the drum and turning on the vibrating axle shaft. The vibrating axle and drum shall be equipped with either antifriction bearings or bronze bushings or graphite impregnated phenolic resin laminated bushings. Appropriate lubrication points shall be provided. The drum support bearings shall have adequate thrust capacity for operation under all field conditions.

3.10 Cleaners. Replaceable cleaners shall be provided over the full width of the drum and shall prevent accumulation of dirt on the drum when the roller is towed in both the forward or reverse direction. The cleaners shall be adjustable to provide a minimum clearance of 3/8 inch between drum and cleaners. Since this clearance will permit dirt to pass between drum and cleaners, the cleaners shall incorporate provisions to prevent accumulation of dirt above the cleaners.

3.11 Rear clevis(s). A rear clevis shall be provided and positioned on the longitudinal centerline of the roller. Unless otherwise specified the clevis shall be of such size to accommodate a lunette as specified in 3.7.1.

3.12 Power unit. The roller shall be powered by a commercial diesel engine having horsepower, torque, and speed characteristics to satisfactorily meet all performance requirements specified herein. The engine shall be equipped with a 24 volt electrical cranking system. The electric cranking system shall include but not be limited to the following: an electric starter, alternator, voltage regulator, battery and all other accessories required to complete the cranking system. The engine shall be mounted as to prevent damage to the engine and attached engine external accessories by load sling cables, when the roller is being transported as a slung load from helicopters specified. The engine shall be equipped with an instrument panel. The panel shall be located in a protected enclosure and shall be isolated so as to dampen the vibration effects. The engine shall be provided with the following accessories:

a. Ammeter or voltmeter.
b. Oil pressure indicator.
c. Oil filter.
d. Air cleaner, dry type.
e. Throttle control.
f. Coolant temperature gage (as applicable).
g. High temperature cut off switch.
h. Low oil pressure cut off switch.
i. Hourmeter-tachometer (combination unit).

The engine shall be capable of operating continuously at rated load in an ambient temperature of 120° F without exceeding the maximum allowable best temperatures established by the engine manufacturer. Certification to this requirement is acceptable (see 4.5.2.3).

3.12.1 Fuel tank. A fuel tank conforming to the contractor's commercial standards, except as specified herein, shall be provided. The tank capacity shall be sufficient for 8 hours of operation at the maximum fuel rating of the

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engine. The tank shall be provided with fuel tank cap gasket, filler neck, and fuel tank cap conforming to MS35643, MS35644, and MS35645 respectively. The tank shall be stenciled: "FILL WITH DIESEL FUEL ONLY". The tank shall be provided with a fuel level indicator and with flexible fuel lines to the engine. Provisions shall be made to drain the fuel into a hand held container.

3.12.2 Engine fuel. The diesel engine furnished shall operate on fuel oil conforming to VV-F-800, grade DF2.

3.12.3 Engine lubrication. The engine lubricating oil shall be in accordance with MIL-L-2104.

3.12.3.1 Engine oil sampling valve. An oil sampling valve shall be provided on the engine. The valve shall be manually operated and shall automatically close after release. It shall be made of material resistant to corrosion such that it will not contaminate the sample. The discharge port of the valve shall be covered with a captive chained cap conforming to SAE J514. The valve shall be located in such a way as to insure that personnel will not be exposed to danger when taking oil samples with the engine running. The location of the oil tap shall be such that when samples are taken, it shall be a true representation of oil that is flowing while the engine is running.

3.12.4 Clutch. When a gear or belt driven vibrating mechanism is provided, a hand-operated, over-center, engagement-type clutch shall be furnished. The clutch shall have a load capacity rating of not less than the maximum energy torque.

3.12.5 Remote controls. When specified (see 6.2), a remote control system shall be provided. The remote control system shall enable the operator to start, stop, and control engine speed, engage and disengage the drum vibrator from the operator's position of a crawler tractor conforming to MIL-T-52510. Provisions shall be provided to secure the controls at the operator's station of the crawler tractor during operation. Provisions shall be provided on the roller for storage of the remote control system.

3.12.6 Battery and battery box.

3.12.6.1 Battery(s). The battery(s) shall be heavy duty automotive type and shall be mounted in a battery box as specified in 3.12.6.2. The battery(s) shall furnish the power for all electrical demands. The battery(s) shall be shipped charged, dry and sealed to prevent entry of moisture during shipment and storage.

3.12.6.2 Battery box. The battery box shall be made of steel having a minimum thickness of 0.0747 inch. The battery box shall be of sufficient size to accomodate Military battery(s) conforming to MS35000. The battery box shall have a lid with an overlapping lip to prevent entry of rain and dirt. The lid shall be provided with not less than two overcenter locking or spring tensioned fasteners to secure the lid in the closed position. The battery box shall be equipped with built-in restraints so constructed that pressure will be applied to the outer battery case only. The battery box shall be mounted securely to the roller and shall be isolated to dampen the vibration of the roller from the batteries and

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battery box. The battery box shall have holes in the bottom for drainage of accumulated moisture. The battery cable openings shall be provided with grommets conforming to MS35489, size as required. The battery cables and grommets shall be secured to the battery box to prevent abrasive cutting by the battery box edge due to vibration. The battery box and terminals shall be coated with acid resisting compounds. The battery terminals shall be provided with guards to prevent accidental shorting and shall be labeled with a "-" in black on the negative terminal and a "+" in red on the positive terminal.

3.12.7 Air cleaner indicator. The dry-type air cleaner shall be provided with an air flow restriction indicator conforming to MS53063.

3.12.8 Slaving components. The roller shall be equipped with a 24-volt receptacle conforming to MS52131. The slave receptacle shall permit charging of the batteries and starting of the engine from an external power source for charging and slaving other equipment. The slave receptacle shall be exterior of the roller near the battery box and shall be labeled: SLAVE RECEPTACLE, 24 VOLTS.

3.13 Vibrating mechanism. A vibrating mechanism shall be provided and shall develop the centrifugal force specified in table I, within the frequency range of not less than 1100 vibrations per minute and not exceeding 2400 vpm. The frequency of the vibrating mechanism shall be variable by not less than 300 vpm within the frequency range specified herein. The vibrations shall be developed by an eccentric shaft, by eccentric weights, or by rotating balls. Gear drive or V-belt drive or hydraulic drive shall be furnished for the vibrating mechanism.

3.13.1 Gear drive. The gear drive shall include all the required gear-boxes, flexible couplings, universal drive shafts and all other necessary mechanical hardware required to provide a gear drive power train having the mechanical integrity to provide optimum operational characteristics and capabilities to operate the roller as specified in 3.1.1.

3.13.2 V-belt drive. V-belt drives shall be in accordance with ZZ-B-225. All belt drives shall be matched multiple V-belt assemblies. Provisions shall be provided to adjust belt tension. Applying tension by use of an idler will be permitted providing the idler is used on the slack side of the belt.

3.13.3 Drive lines. All drive lines shall be installed in such a manner so as to prevent damage to drive line components by load slings during transportation by helicopters organic to military airmobile operations employing slung load transportation methods.

3.13.4 Hydraulic drive. The hydraulic drive shall include the required hydraulic pump(s), motor(s), lines, control valve(s), reservoir, filter(s) and all other necessary hardwear required to provide a hydraulic drive having the operational characteristics and capabilities to operate the roller as specified in 3.1.1. All of the hydraulic components shall be compatible with oil conforming to MIL-L-2104, grade 10.

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3.14 Identification marking. The roller shall be identified in accordance with MIL-STD-130. The marking shall be applied to the roller on plates conforming to MIL-P-514, type I, style 1, composition C, grade A, class 1 material. The plates shall be attached by screws, bolts, or rivets in a conspicuous location. The identification plate(s) shall show the model number, NSN, serial number, weight, date of manufacture and date of delivery.

3.14.1 Labeling. Labels and instructions shall be mechanically attached to the roller. Adhesives shall not be used. Valve controlls shall be provided with double-ended arrows showing direction of operations and labeled at each end to indicate the functional result (e.g. open, close, etc.).

3.15 Shipping data plate. Shipping data plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material and in addition shall show the silhouette of the roller in the transport position indicating the center of gravity and the location and capacity of the lifting and tiedown attachments. The plates shall be attached by rivets, screws, or bolts, in conspicuous protected locations.

3.16 Instruction plate. An instruction plate, conforming to MIL-P-514, type III, composition C, of type I, grade A, class 1 material shall contain clear and concise instructions for operation of all controls and for engine start. The plate shall be permanently affixed to the roller in a conspicuous place.

3.17 Electromagnetic interference. Unless otherwise specified (see 6.2), the electromagnetic interference emission and susceptibility characteristics of the roller shall conform to SAE J551, devices category.

3.18 Lubrication. All surfaces that require lubrication shall be provided with an accessible means for lubricating.

3.18.1 Lubricants. The procedure for the selection of lubricants shall be in accordance with MIL-STD-838, section 5. Lubricants selected shall be in accordance with MIL-HDBK-113, chapter 2. When the specification of the lubricant selected includes a requirement for a qualified products list (QPL), the lubricant supplied shall be from a source that is listed on the applicable QPL.

3.18.2 Lubricating fittings. Lubrication fittings shall conform to SAE J534. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.18.3 Pressure-release device. An automatic pressure-release device shall be provided where the use of pressure-lubricating equipment will damage grease seals or other parts.

3.18.4 Filling, draining, and checking provisions. Enclosures such as gearcases that contain a reservoir of lubricants shall be equipped with dipsticks, check plugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped for

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filling the enclosure with lubricant and for draining. The drain shall be fitted with a magnetic drainplug. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be accomplished when the roller is in its normal operating position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to a hand held container.

3.18.5 Grease lubrication. All grease lubrication including lubrication of sealed bearings, shall be with greases conforming to MIL-G-10924 or MIL-G-23827 as applicable. The roller shall be assembled, run-in, tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because military lubricants may not be compatible with other greases. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

3.18.6 Oil lubrication. Unless otherwise specified herein, all components shall be filled to the operating level with military oils designed for use in the temperature range as specified herein. A tag shall be attached in a conspicuous place to indicate the temperature range and grade of lubricant used. The high side bearings shall not be starved when the roller is operated on a 10 degree slope determined around the longitudinal axis of the roller.

3.19 Fungus and moisture resistance. Unless otherwise specified (see 6.2), the electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I, with 1 percent copper 8-quinolinolate (by weight) or 7 percent salicylanilide based on the nonvolatile content of varnish:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.20 Transportability. The roller shall be capable of being transported by highway, rail, water, and C130, C141, C5 aircraft and helicopters. The roller shall withstand the impact forces encountered in air and rail shipments without damage or permanent deformation. The roller shall conform to MIL-STD-669 for low velocity airdrop and low altitude parachute extraction (LAPES) from C-130 and C-141 aircraft when tested in accordance with 4.5.2.8. When rigged for airdrop in accordance with MIL-STD-669, the roller shall be capable of being transported aboard C-130 and C-141 aircraft in accordance with MIL-A-8421 and shall meet the established tip-off curves for C-130 (see figure 3) and C-141 (see figure 4) aircraft. The roller shall be equipped with slinging provisions and shall be capable of being lifted without damage using slings. The roller shall be equipped with tiedown provisions for transport by the above modes.

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Suspension and extraction parachute provisions shall be provided for parachute delivery. Complete diagrams and instructions for lifting the roller and tying down the roller on freight cars and truck trailers shall be furnished in the operator's manual. The operator's manual shall also include instructions for component removal when required for transport.

3.20.1 Parachute suspension provisions. Four integral suspension provisions conforming to MIL-STD-814 shall be furnished on the roller.

3.20.2 Slinging provisions. The roller shall be provided with four slinging provisions conforming to MIL-STD-209. The provisions shall enable the roller to be lifted in the normal traveling or operating position. Provisions shall be positioned such that the plane of each provision is in approximate alignment with the anticipated angle of the sling leg. Parachute suspension provisions may be used as slinging provisions if they meet the requirements of MIL-STD-209.

3.20.3 Tiedown provisions. The roller shall be provided with integral provisions to permit tiedown of the roller to the floor or deck of the transportation medium or the airdrop platform. The tiedown provisions shall conform to MIL-STD-209, MIL-STD-814, and MIL-A-8421.

3.20.4 Extraction parachute provision. A single fitting shall be provided on the roller for attachment of the parachute extraction system. The fitting shall conform to MIL-STD-814.

3.20.5 Rail transportability. The roller shall be capable of withstanding the forces imposed during rail impact testing in accordance with MIL-STD-810.

3.20.6 Air transportability. The roller shall be capable of being loaded, unloaded, and transported aboard C-130, C-141, and C-5 aircraft in accordance with MIL-A-8421. The roller shall be capable of being externally transported by CH-47 and larger cargo helicopters. The roller (type I only) shall also be capable of being internally transported by CH-47 cargo helicopters.

3.21 Treatment and painting. The portions of the roller normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2) top coat color shall be camouflage green 383 conforming to MIL-C-46168. The roller shall then be overcoated in accordance with the Government furnished camouflage patterns with MIL-C-46168 top coat of the colors specified in the camouflage patterns, with the exception that the first article may be camouflage green 383.

3.21.1 Data for camouflage patterns. The contractor shall generate the following data (see 6.2).

a. Eight by ten color photos that mirror each of the views listed under 3.21.2.
b. Original illustrations of each of the views listed under 3.21.2. The illustrations shall be to 1/8 scale on D size paper. Each illustration shall include length, width, and height dimensions relative to each

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other and shall be clearly detailed to the extent that all discontinuities of the spreader that cover one square inch or more of area, are clearly delineated to scale.

3.21.2 Camouflage pattern views.

a. Back
b. Front
c. Left side
d. Right side
e. Top
f. Hidden views that would result in a change of a finished pattern.

NOTE: Hidden views are those areas requiring camouflage coloration that are not apparent from the other views. Included are variations in the appearance of the spreader that result when the deployed spreader is taken from a stand-by condition and placed into a ready-for-use condition.

3.21.3 Government-furnished property. The following property will be furnished by the Government (see 6.5).

Item No.	Description	Identification	Quantity
1	Pattern drawings	Camouflage patterns	As required

3.22 Stenciling. The gross weight of each roller shall be stenciled on each side of the roller. "Lift" and "Tiedown", as applicable shall be stenciled adjacent to their respective tiedown and lifting attachment. All stenciling shall be in block-type letters not less than 1 inch high and shall be lusterless black in color.

3.23 Toolbox. A waterproof toolbox made of steel not less than 0.0747 inch nominal thickness shall be provided and shall be of sufficient size to hold those tools and accessories necessary to operate the roller. The toolbox shall have a hinged lid with the hinge extending the full length of one edge and shall be provided with not less than two fasteners that will keep the lid closed when the toolbox is subjected to vibration. The toolbox lid shall have an overlapping lip to prevent entry of rain and dirt. The toolbox shall be securely mounted in a protected and accessible location permitting the lid to be opened through an arc of at least 90 degrees. The lid shall remain in the fully open position by the weight of the lid or by holding devices. The toolbox lid shall be provided with a hasp that will accommodate a padlock that has a 3/8 inch diameter shackle. The toolbox shall have a hole in the bottom for drainage of excess moisture.

3.24 Workmanship. The roller shall be free from defects such as incomplete welds, rust, cracks, and other defects that could impair the operation of the spreader. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using

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tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.24.1 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping, wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the spreader to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch, unless appropriate bridging techniques are used. Unless otherwise specified, the welding process used in fabrication of the roller shall be at the option of the contractor (see 6.2).

3.24.1.1 Welder certification. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by AWS B2.1. The certification shall be made available for review by the contracting officer or designated representative.

3.24.2 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.24.3 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a

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part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 Examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The first article shall be tested as specified in 4.5.2.1 through 4.5.2.13. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each roller shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Test. Each roller shall be tested as specified in 4.5.2.13. Failure of the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The roller shall be examined for the following defects:

101. Assembly not complete or not correct.
102. Dimensions and weight not as specified.
103. Material not as specified.
104. Appearance of rust and corrosion.
105. Materials not corrosion resistant.
106. Dissimilar metals not in accordance with MIL-STD-889.
107. Material and or material finishes information not available.
108. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the roller.
109. Safety devices, guards, insulation, not as specified.
110. Maintainability not as specified.
111. Parts interchangeability not as specified.
112. Frame not as specified.
113. Drawbar not as specified.
114. Support leg not as specified.
115. Drum not as specified.

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116. Axles and bearings not as specified.
117. Drum cleaners not as specified.
118. Rear clevis(s) not as specified.
119. Power unit or power unit accessories not as specified.
120. Fuel tank not as specified.
121. Diesel engine furnished does not operate on fuel oil specified.
122. Engine lubricating oil not as specified.
123. Engine oil sampling valve not as specified.
124. Clutch not as specified.
125. Remote control (when furnished) not as specified.
126. Battery or battery box not as specified.
127. Air cleaner not as specified.
128. Slaving components not as specified.
129. Vibrating mechanism not as specified.
130. Drive system not as specified.
131. Identification marking missing or not as specified.
132. Shipping data plates missing or not as specified.
133. Instruction plate missing or not as specified.
134. Grease or oil lubricants not as specified.
135. Lubrication fittings not accessible, missing, or not installed as specified.
136. Pressure-release device not as specified.
137. Filling, draining, and checking provisions not as specified.
138. Fungus and moisture resistance not as specified.
139. Lifting, parachute suspension, extraction parachute, and tiedown attachments not as specified.
140. Treatment and painting not as specified.
141. Color not as specified.
142. Camouflage pattern data not as specified.
143. Camouflage patterns not as specified.
144. Stenciling not as specified.
145. Toolbox not as specified.
146. Workmanship not as specified.
147. Welding not as specified.
148. Welders' certification not made available.
149. Bolted connections not as specified.
150. Riveted connections not as specified.
151. Human factors not as specified in 3.6.

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The roller shall operate as specified herein without maintenance other than the contractor's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to test. Prior to testing, lubricate the roller with military lubricants as specified in 3.18. After checking and servicing the roller, connect the lunette of the roller drawbar to the drawbar or pintle of the test towing unit. When a remote control system is furnished (see 3.12.5 and 6.2) the test specified in 4.5.2.2 shall be conducted using the remote control system to control the roller operation.

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4.5.2.2 Towing. Start the engine using the engine controls and engage the vibrating mechanism. Tow the roller over not less than 6 inches (type I) or 12 inches (type II) of loose dirt fill or plowed earth at a speed of not less than 1-1/2 miles per hour and not more than 3 miles per hour for 10 successive passes. One pass shall consist of not less than 100 linear yards. Repeat this test for 20 operating hours. Stop the engine once during each 2-hour operating period and restart it after a 5-minute shutdown period. During the operational test determine the VPM frequency range.

4.5.2.2.1 Failure criteria. Inability to start the engine using only the engine controls, inability to engage or disengage the vibrating mechanism, inability of the cleaners to prevent accumulation of dirt as specified in 3.10, inability to stop or restart the engine using only the engine control system, leakage of fuel or lubricants, or permanent deformation or damage shall constitute failure of this test. Inability to vary the VPM frequency range as specified (see 3.13) shall constitute failure of this test.

4.5.2.3 High temperature operation. The contractor shall furnish test data to prove that the roller is capable of operating in an ambient temperature of 120° F without exceeding the engine manufacturer's recommended limits for radiator top tank temperature of applicable engine sump and gallery oil temperatures and hydraulic component manufacturer's hydraulic fluid temperature limits.

4.5.2.3.1 Failure criteria. Failure of the contractor to provide test data or manufacturer's certification (see 3.12) shall constitute failure of this test.

4.5.2.4 Parachute suspension provisions. The parachute suspension provisions shall be demonstrated to prove conformance to 3.20.1. Apply a force of 1.5 times the working load as specified in MIL-STD-814 to each of the parachute suspension provisions. Permanent deformation or failure of any suspension provision, connection, or structural member of the roller shall constitute failure of this demonstration.

4.5.2.5 Slinging provisions. The slinging provisions shall be demonstrated to prove conformance to 3.20.2. Permanent deformation or failure of any slinging provision, provision connection, or structural member of the roller shall constitute failure of this demonstration. The demonstration procedure shall be as follows:

a. Lift the roller and hold in suspension in its normal travel position using slings that converge not more than 24 feet above the lowest extremity of the roller. Spreader bars are permitted. Determine the force and angles of application for each provision. Measure the clearance between each sling and the roller.
b. Restrain the roller by anchoring the main frame and subject each slinging provision to a force equal to the working load specified in MIL-STD-209. This force shall be applied in the direction as determined in a. above. Hold each load for a minimum of 90 seconds.

4.5.2.6 Tiedown provisions. The tiedown provisions shall be demonstrated to prove conformance to 3.20.3. Apply lateral and vertical forces to each tiedown

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provision as specified in MIL-STD-814. Apply a longitudinal force to each tiedown provision as specified in MIL-STD-209. Permanent deformation or failure of any tiedown provision, provision connection, or structural member of the roller shall constitute failure of this demonstration.

4.5.2.7 Extraction parachute provision. The extraction parachute provision shall be demonstrated to prove conformance to 3.20.4. Apply a force equal to 1.5 times the working load as specified in MIL-STD-814. Permanent deformation or failure of the extraction provision, provision connection, or structural members of the roller shall constitute failure of this demonstration.

4.5.2.8 Airdrop. The roller shall be subjected to simulated airdrop impact demonstration in accordance with MIL-STD-669 to prove conformance to 3.20. This demonstration will be conducted by the Government at a Government facility; however, acceptance of the first article will be dependent on the successful completion of this demonstration. The demonstration shall consist of the following:

a. Rig the roller in accordance with MIL-STD-669 and confirm that the rigged weight and dimensions conform to the requirements of MIL-A-8421 and the tip-off curves for C-130 (see figure 3) and C-141 (see figure 4) aircraft.
b. Drop test the roller in accordance with MIL-STD-669.
c. After the first drop, the roller shall be de-rigged, examined, started, and driven forward and backward a few feet, and all controls shall be actuated.

Failure of the rigged roller to meet the weight and dimensional requirements of MIL-A-8421 or the tip-off curves (see figures 3 and 4) permanent deformation or failure of any provision connection or structural member, any structural damage to the roller, or inability of the roller to perform its intended functions shall constitute failure of this demonstration.

4.5.2.9 Rail impact. The roller shall be rail impact tested in accordance with MIL-STD-810, method 516.3, procedure VIII, using a conventional tiedown procedure. Permanent deformation or failure of any tiedown provision, provision connection, structural member, or any structural damage to the roller or inability of the roller to perform its intended function shall constitute failure of this demonstration.

4.5.2.10 Helicopter lift. The roller shall be subjected to an external sling-lift by CH-47 or larger cargo helicopter. This demonstration will be conducted by the Government at a government facility; however, acceptance of the first article will be dependent on the successful completion of this demonstration. Permanent deformation or failure of any slinging provision, provision connection or structural member, any structural damage to the roller, or inability of the roller to perform its intended function shall constitute failure of this demonstration.

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4.5.2.11 Electromagnetic interference. Unless otherwise specified, the roller shall be tested in accordance with SAE J551. A report tabulating the allowable maximum interference and the measured interference shall be furnished with the first article test report.

4.5.2.11.1 Failure criteria. Failure to meet the requirements of SAE J551 shall constitute failure of this test.

4.5.2.12 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum: noise levels shall be measured when equipment is operating under full load, MIL-STD-1474, 5.1.2.1.4. The noise level at the typical operation position shall be provided as dB(a) level. Failure to comply with MIL-STD-1474 provisions and 3.5.2.1 shall constitute failure of this test.

4.5.2.13 Quality conformance operational run-in. Each roller shall be serviced as required and shall be run-in in accordance with the manufacturer's normal procedures for a period of not less than 60 minutes.

4.5.2.13.1 Failure criteria. One or more functional defects shall constitute failure. Acceptance of the equipment shall not be approved until all defects are corrected.

4.6 Inspection comparison. The Government may select rollers at any time during the contract production period and subject these rollers to the examination specified in 4.5.1 and the tests specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government, and will not include the previously inspected first article rollers. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison roller to meet any requirements specified herein during and as a result of the the examination and test specified in 4.6 shall be cause for rejection of the inspection comparison roller(s) and shall be cause for refusal by the Government to continue acceptance of the production rollers until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on rollers previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all rollers accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

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4.7 Inspection of packaging. The preservation, packing, and marking shall be examined in accordance with the applicable quality assurance provisions of MIL-R-3075.

5. PACKAGING

5.1 Preservation, packing and marking. Each complete roller shall be preserved, packed and marked in accordance with the applicable requirements of MIL-R-3075. The degree of preservation shall be level A, or commercial and the degree of packing shall be level A, or commercial, as specified (see 6.2). Commercial preservation, packing, and marking shall be in accordance with ASTM D 3951.

6. NOTES

6.1 Intended use. The rollers are intended for compaction of earth fill, gravel, and soil stabilization areas. The rollers are intended to be air transportable within the range of an air transportable division.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type of roller required (see 1.2).
c. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
d. Time frame for submission of first article roller and number of rollers required (see 3.2).
e. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests will be conducted by the contractor (see 3.2).
f. When first article roller inspection is not required (see 3.2).
g. When draw bar shall be compatible with a crawler tractor drawbar conforming to SAE J749a pin size 1.75 (see 3.7.2).
h. When remote control system is required (see 3.12.5).
i. When electromagnetic interference requirements are other than as specified (see 3.17).
j. When fungus and moisture resistance is not required (see 3.19).
k. When the finishing coat shall be other than specified (see 3.21).
l. Time frame when camouflage pattern data is required (see 3.21.1).
m. When a welding process other than as specified is required (see 3.24.1).
n. Degree of preservation and degree of packing required (see 5.1).

6.2.1 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, contract data requirements list (CDRL), the data requirements identified below shall be developed as specified by an approved data item description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410.6 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance

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with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirements	Applicable DID No.
a. 3.21.1	Camouflage pattern photos	UDI-V-24003B/M (see figure 1)
b. 3.21.1	Camouflage pattern illustrations	DI-M-30420/M (see figure 2)

6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of one roller. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the document's first article.

6.4 Lubricants. MIL-STD-838 Lubrication of Military Equipment, prescribes the policy for using specification-type products wherever possible and provides specific requirements for potential use of non-standard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113 Guide for the Selection of Lubricants, Fluids, Preservative and Speciality Products for Use in Ground Equipment Systems. The contracting officer should note that unless otherwise authorized by the US Army Belvoir Research & Development Center (ATTN: STRBE-VF), Fort Belvoir, VA 22060-5606, lubricants, fluids, and greases for ground equipment systems must be restricted to those listed under MIL-HDBK-113, chapter 2.

6.5 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.21.3.

6.6 Degrees of packing have not been specified as the packing specified is applicable for all degrees of packing. For the purposes of preservation/packing level marking the marking shall be level A.

Custodian:
Army - ME

Review activity:
Army - MT

Preparing activity:
Army - ME

Project 3895-A228

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DATA ITEM DESCRIPTION	2. IDENTIFICATION NO(S).	
	AGENCY	NUMBER
1. TITLE PHOTOGRAPHIC PRINTS AND SLIDES	NAVY-YD	UDI-V-24003B/M

3. DESCRIPTION/PURPOSE

This data consists of photographic prints and/or slides which provide a pictorial medium for exchange of technical information between developing Agency and users and becomes a permanent historical and legal record by recording configuration during the course of program development.

4. APPROVAL DATE
13 June 1972

5. OFFICE OF PRIMARY RESPONSIBILITY
NAVFAC 0434

6. DDC REQUIRED

8. APPROVAL LIMITATION

7. APPLICATION/INTERRELATIONSHIP

7.1 Deliverable data covered by this DID and specified in block 16 of the CDRL, is required as a visual record of equipment procured under contract.

9. REFERENCES (Mandatory as cited in block 10)

MCSL NUMBER(S)

10. PREPARATION INSTRUCTIONS

10.1 These photographs shall be standard 8x10 inch prints and furnished in 2x2 inch cardboard frames; and unless otherwise specified on the DD Form 1423. Views shall include, but not be limited to, a front, side, and rear view. Background of photographs shall be void of unrelated objects.

10.2 Interior views shall be in sufficient quantity to show arrangement and layout of installed equipment.

10.3 Photographic prints shall show excellent detail and shall be color suitable for reproduction.

DD FORM 1664

FIGURE 1. Camouflage pattern photos.

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DATA ITEM DESCRIPTION	2. IDENTIFICATION NO(S).	
	AGENCY	NUMBER
1. TITLE Technical Illustrations and Graphs	USAF	DI-M-30420/M
3. DESCRIPTION/PURPOSE To provide wave plotting, drafting, computations, data reduction and/or mathematical analysis in support of research projects.	4. APPROVAL DATE 29 Oct 1976	
	5. OFFICE OF PRIMARY RESPONSIBILITY AFSC	
	6. DDC REQUIRED	
	8. APPROVAL LIMITATION	
7. APPLICATION/INTERRELATIONSHIP Mathematical Services Projects	9. REFERENCES (Mandatory as cited in block 10)	
Replaces (U)M-177/ESD	MCSL NUMBER(S)	

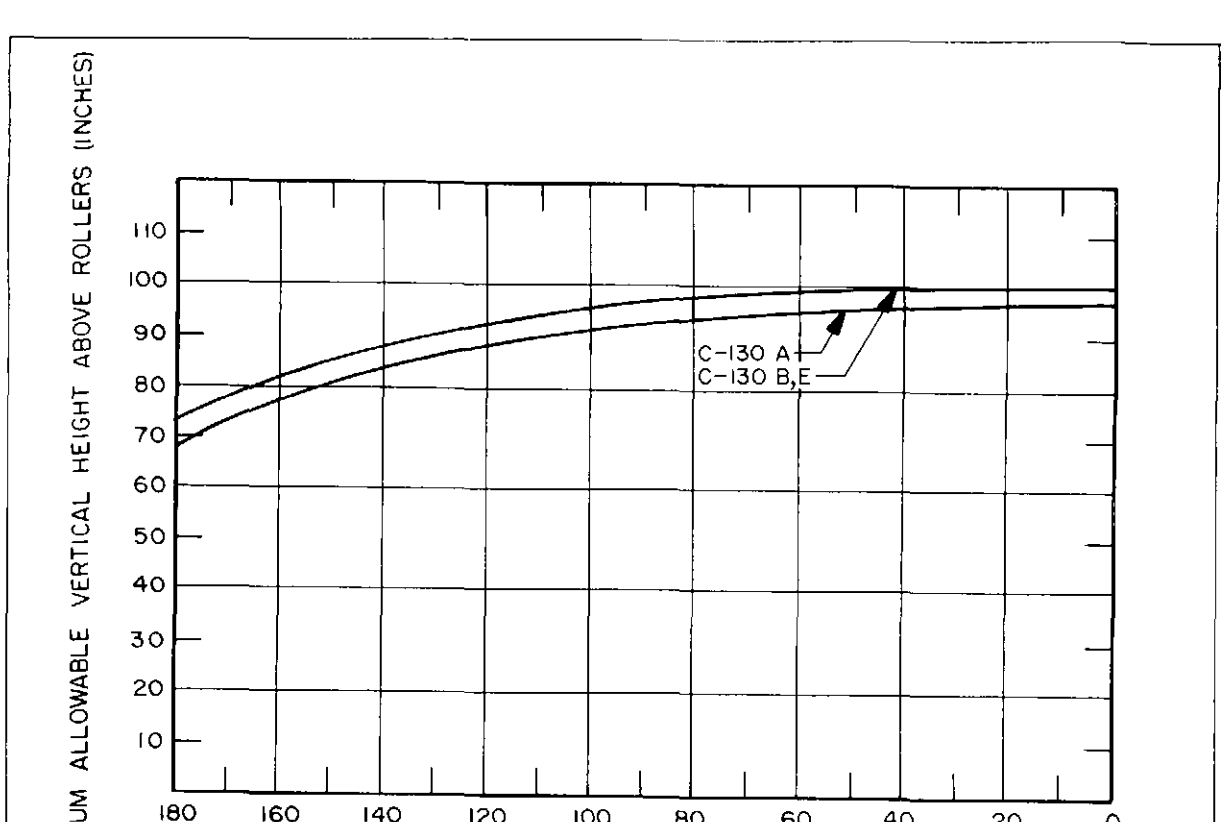
10. PREPARATION INSTRUCTIONS

Data will encompass plotting, drafting and artistic creative illustrations of apparatus and devices. The work may include three-dimensional pictorial representations. Completed reproducible art work must be of sufficiently high quality to be used for publication in technical journals and for visual aids at conferences, reviews and briefings.

DD FORM 1664

FIGURE 2. Camouflage pattern illustrations.

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METRIC EQUIVALENTS	
INCHES	METERS
10	0.254
20	0.508
30	0.762
40	1.02
50	1.27
60	1.52
70	1.78
80	2.03
90	2.29
100	2.54
110	2.79
120	3.05
140	3.56
160	4.06
180	4.57

FIGURE 3. C-130 TIPOFF CURVE

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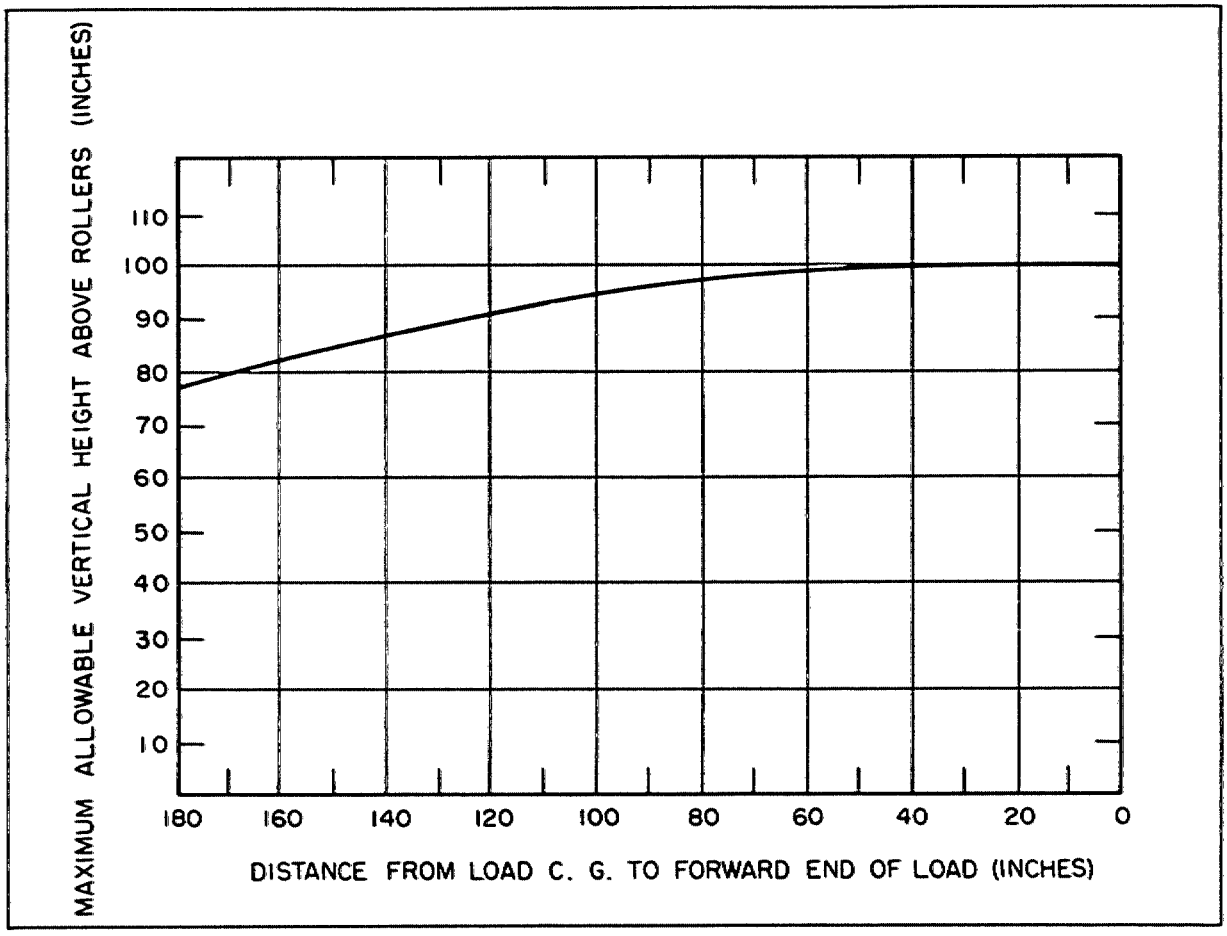


FIGURE 4. C-141 TIPOFF CURVE

X-4501

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-0052499D(ME)	Rollers, Towed, Smooth Drum, Vibratory, Airborne/Airmobile, Single-Drum

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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